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How experience shapes startup screening: cognitive strategies of startup accelerator evaluators

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ABSTRACT

Accelerators screen large numbers of early-stage ventures under f uncertainty, limited information, and time pressure. Yet, little is known about how evaluative experience shapes the cognitive processes underlying initial screening decisions. Using Verbal Protocol Analysis (VPA), this exploratory study captures evaluators' concurrent verbalizations as they assess startup applications, enabling a comparison of how novice and experienced evaluators process information in real time. The findings reveal systematic differences in screening cognition. Experienced evaluators reached decisions more quickly and focused selectively on team quality, market plausibility, and accelerator fit. In contrast, novices engaged more extensively with information explicitly provided in applications, particularly product and financial details. Heuristic reasoning also varied by experience. Experienced evaluators more frequently drew on prior screening cases and domain-specific knowledge, whereas novices relied more heavily on generalized prototypes and personally salient experiences. Overall, the results suggest that evaluative expertise lies less in the quantity of knowledge available to evaluators and more in how information is filtered, prioritized, and interpreted under uncertainty. The study contributes to accelerator selection and experiential learning research by opening the cognitive "black box" of initial screening and highlighting expertise as a context-sensitive process of selective information processing. Practical implications for evaluator development are discussed.

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1. Introduction

Entrepreneurs face significant challenges in transforming early-stage ideas into market-ready ventures (Le and Suh 2019; Marcon and Ribeiro 2021; Mungila Hillemane, Satyanarayana, and Chandrashekar 2019). Accelerator programs emerged to support and accelerate this process by providing valuable learning and networking opportunities (Cohen, Bingham, and Hallen 2019; Pauwels et al. 2016) and ultimately enhancing the legitimacy of participating startups (Yin and Luo 2018). However, the success of acceleration programs depends, in part, on their ability to select a portfolio of high-potential

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startups (Pauwels et al. 2016; Yin and Luo 2018). The screening and selection of potential accelerated businesses take place under high levels of uncertainty, requiring evaluators to make judgments based on limited information. The novelty of the ideas, the lack of validation of revenue models, and information gaps further complicate the assessment (Wright, Koning, and Khanna 2023). These challenges are particularly pronounced during the initial screening phase, when large numbers of applicants must be evaluated using brief pitches or short written descriptions, making in-depth due diligence impractical (Wright, Koning, and Khanna 2023). Under these conditions, evaluators must assess venture potential in the absence of reliable performance information (Cohen, Bingham, and Hallen 2019; Leatherbee and Gonzalez-Uribe 2018). Yet, despite the central role of evaluation in this process, research has paid limited attention to the internal mechanisms and cognitive processes through which accelerator screening judgments are formed (Butz and Mrożewski 2021; Leatherbee and Gonzalez-Uribe 2018).

Under such conditions, evaluators cannot optimally consider every available piece of information. Instead, they must selectively attend to, filter, and interpret signals that appear relevant to the screening decision. Research on decision-making under uncertainty suggests that decision-makers often rely on simplified decision strategies when exhaustive information processing is difficult or impractical. In such contexts, experienced decision-makers adapt their information processing to the structure of the task environment and rely on selective cues when making judgments (Luan, Reb, and Gigerenzer 2019). Thus, the key question is not simply which startups evaluators select, but how evaluative experience may change the way they filter, prioritize, and interpret the signals available during initial screening.

Research on early-stage investment decision-making provides important insights into this question. Venture capitalists and business angels literature suggest that repeated exposure to investment decisions can shape evaluators' judgment and information-processing strategies (T. Botelho, Harrison, and Mason 2021; De Clercq and Sapienza 2005; Harrison, Mason, and Smith 2015). For example, Harrison, Mason, and Smith (2015) found differences between business angels with different levels of investment experience in decision speed, the emphasis placed on investment criteria, and the use of experience-based heuristics.

Although accelerator screening shares important characteristics with business angel and venture capital evaluation, it represents a distinctive decision context. Evaluators must process a large volume of applications within a time-bounded cohort cycle, make judgments about very early-stage ventures on the basis of limited and often uncertain information, and determine which ventures warrant more intensive evaluation in the subsequent selection stage (Cohen, Bingham, and Hallen 2019; Roberts and Lall 2018; Woolley and MacGregor 2022). Accordingly, evaluators may consider not only a venture's current potential but also its potential to develop through the acceleration process. However, limited research has examined how evaluative experience shapes these cognitive processes in accelerator settings, making initial accelerator screening a particularly relevant context for studying the development of evaluative expertise under uncertainty. In this exploratory study, we examine how experience shapes the decision-making processes of novice and experienced accelerator evaluators, focusing on their evaluation criteria, cognitive processes, and allocation of attention during startup screening. To address this, we employ Verbal Protocol Analysis (VPA), capturing evaluators' real-time

decision-making by collecting and analysing the verbalizations made by individuals as they perform a task (Ericsson and Simon 1993).

The study makes two primary contributions. First, it contributes to research on accelerator selection by opening the cognitive black box of initial screening, shifting attention from the criteria evaluators use to how they process and interpret application information in real time. Second, the study contributes to experiential learning and expertise research by showing that evaluative experience may be reflected not simply in accumulated knowledge but also in changes in cognitive processing patterns. The study examines whether experience accumulated through accelerator screening is associated with differences in the pace and selectivity of information processing, attention to cues, and reliance on prior knowledge when forming screening judgments. Together, these contributions highlight evaluative experience as a potential source of variation in screening cognition.

The remainder of the paper is structured as follows. First, we review the literature on startup accelerators and their selection processes, highlighting gaps that motivate this study. Next, we describe the research methodology, including data collection and analytical procedures. We then present the findings, organized around key emergent themes, and discuss their implications for theory and practice within the broader entrepreneurship literature. Finally, we outline the study's limitations and propose directions for future research in this rapidly evolving field.

2. Literature review

2.1. Startup accelerators

Startup accelerators are organizations that provide early-stage ventures with intensive, time-limited support programs that combine mentorship, training, networking, and seed funding in exchange for equity (Dempwolf, Auer, and D'ippolito 2014; Svetek 2022). Since their emergence in the mid-2000s, accelerators have developed a distinctive model characterized by cohort-based participation and close interaction with experienced mentors and investors (Moroz, Sierra, and Anderson 2024). Programs typically last between three and six months and are designed to accelerate ventures from ideas to investment readiness (Assenova and Amit 2024).

Through structured feedback cycles and frequent performance evaluations, accelerators encourage founders to translate learning and feedback into tangible outputs – including prototypes, validated business models, and investor-ready pitch materials (Hallen, Cohen, and Bingham 2020). This intensive format supports the development of practical managerial skills and market-relevant knowledge, enabling participants to progress more rapidly than in less structured environments (Assenova and Amit 2024). Accelerators therefore function not only as sources of early-stage finance but also as platforms for learning and venture development.

Accelerators occupy a hybrid position in the entrepreneurial finance ecosystem, combining elements of traditional investors with those of startup support organizations (Berliner, Shneor, and Wald 2025). From the supply side, they introduce investment logics that complement venture capital and angel investment (Butticè and Vismara 2022). From the demand side, they expand entrepreneurs' access to early-stage funding, networks and legitimacy, thereby influencing venture governance and growth trajectories (Berliner,

Shneor, and Wald 2025; Eslamloo, Brown, and Amrollahi Biyouki 2025). By facilitating systematic interaction with mentors and investors, accelerators reduce informational asymmetries and address capability gaps that hinder commercialization (Thomas and Georgee 2020).

Empirical research suggests that accelerator participation is associated with positive venture outcomes. Through mentorship and vicarious learning, founders enhance their decision- making capabilities and execution skills (Hallen, Cohen, and Bingham 2020). Accelerated startups are more likely to attract venture capital, generate revenue, expand employment, and offer higher wages than comparable nonparticipants (Assenova and Amit 2024), indicating that accelerators contribute meaningfully to venture performance.

Access to these benefits is governed by a highly selective admission process. Given limited program capacity, accelerators must construct portfolios of high-potential ventures to maximize program effectiveness and investment return (Pauwels et al. 2016; Yin and Luo 2018). Thus, the quality of selection is a key determinant of accelerator success (Mian, Lamine, and Fayolle 2016). Consequently, the judgments made by evaluators during early screening are central in shaping cohort composition and subsequent venture outcomes. Understanding how these investment decisions are formed is therefore critical for advancing knowledge of accelerator effectiveness and serves as the focus of the following section.

2.2. Investment decision-making by accelerators

Early-stage startup evaluation is characterized by substantial uncertainty, limited performance information, and pronounced information asymmetry between founders and potential funders. Most ventures entering accelerator programs lack established revenue streams, validated markets, or stable organizational structures, making traditional financial assessment tools insufficient (Ren, Raghupathi, and Raghupathi 2025). Consequently, predicting which ventures will ultimately succeed is particularly difficult because they are evaluated before reliable indicators of future performance have emerged. Evaluators must therefore form judgments based on information that is incomplete, ambiguous, and subject to continual change.

Moreover, evaluators often face high application volumes and substantial time pressure during the screening process. Under these conditions, fully analytical and exhaustive processing of every piece of available information is rarely feasible. Although formal criteria provide structure to the assessment process, their interpretation depends on evaluators' subjective judgments about the meaning and relative importance of particular signals (Dumont 2024; Lavi and Yaniv 2023). Such judgments are shaped not only by the information provided in venture applications but also by knowledge accumulated through prior experience. Research in cognitive decision-making suggests that individuals draw on and update knowledge derived from previous experiences when making judgments under uncertainty (Daw and Doya 2006; Sutton, Barto, and Barto 1998). Experience is therefore especially relevant to early-stage investment decisions, where evaluators must interpret sparse and ambiguous signals rather than simply verify established indicators of performance (Beyhan, Akçomak, and Cetindamar 2024; Yin and Luo 2018). This is especially salient in accelerator screening, where the scarcity of reliable performance data makes evaluators' experiential knowledge an important basis for assessing ventures'

growth potential (Leatherbee and Gonzalez-Uribe 2018). Understanding how accumulated experience shapes the way evaluators process such information is therefore essential for explaining how screening judgments are formed in accelerators.

Experiential learning theory provides a useful foundation for understanding this process. Kolb (1984) conceptualizes learning as the transformation of experience into knowledge through cycles of concrete experience, reflection, conceptualization, and subsequent experimentation. By explaining how individuals transform repeated experiences into knowledge, experiential learning theory provides a useful basis for understanding how different knowledge structures emerge and develop over time (Holcomb et al. 2009). Through repeated exposure to decision situations, individuals can progressively develop and refine cognitive frameworks that organize subsequent experiences and guide future action (Gabrielsson, Julkunen, and Andersson 2025; Lattacher et al. 2024). In entrepreneurial contexts, accumulated experience has been associated with increasingly developed frameworks that help individuals recognize patterns and connect seemingly unrelated information (Baron and Ensley 2006). Such learning can also create domain-specific path dependencies, whereby prior experience channels attention toward familiar aspects of the decision environment and makes it more efficient to identify and assess cues that have proved informative in that domain (Holcomb et al. 2009). Applied to accelerator screening, this suggests that repeated evaluative experience may gradually reshape not only what evaluators know, but also how they allocate attention, which cues they regard as informative, and how they interpret ambiguous information.

These experience-shaped knowledge structures provide a basis for understanding selective information processing under uncertainty. Under conditions of information overload, uncertainty, and time pressure, evaluators cannot process all available information exhaustively and may therefore rely on simplified decision strategies to reduce the complexity of evaluation (Luan, Reb, and Gigerenzer 2019; Maxwell, Jeffrey, and Lévesque 2011; Zacharakis and Shepherd 2001). Heuristics are consequently not necessarily departures from rational decision-making; they can function as adaptive strategies for interpreting ambiguous signals and prioritizing information that is more likely to be informative (Bingham and Eisenhardt 2011; Gigerenzer, Reb, and Luan 2022). In highly uncertain environments, where the reliability of available information is difficult to establish, focusing on a limited set of informative cues can reduce sensitivity to noisy or less relevant information (Gigerenzer, Reb, and Luan 2022). From this perspective, heuristics can be viewed as one potential cognitive manifestation of accumulated experience: as evaluators learn from repeated screening encounters, prior experiences may become embedded in simplified judgment strategies that guide which cues they notice, how they interpret them, and how quickly they reach a judgment (Harrison, Mason, and Smith 2015). The usefulness of such experience-informed heuristics, however, depends on whether the cues and patterns learned from prior experience remain informative in the current decision environment. Thus, the diagnostic value of a cue is not universal but may vary across venture types, industries, accelerator missions, and broader institutional contexts (Gigerenzer, Reb, and Luan 2022).

Evidence from entrepreneurial finance and expertise research provides support for this experience-based transformation of evaluative cognition. Experienced investors appear to process investment opportunities more selectively, relying more heavily on qualitative and contextual cues such as founder characteristics and team dynamics, whereas less

experienced investors place greater reliance on explicit and readily observable indicators (Bernstein, Korteweg, and Laws 2017; Warnick et al. 2018; Wiltbank et al. 2009). Similar differences have been documented in entrepreneurial opportunity recognition, where experienced entrepreneurs draw on more refined cognitive frameworks to recognize patterns and connect information that may appear unrelated to less experienced decision-makers (Baron and Ensley 2006; Politis 2005). These more developed knowledge structures may also enable experienced decision-makers to recognize subtle patterns and relationships more readily, supporting faster judgments than those of less experienced decision-makers (Luan, Reb, and Gigerenzer 2019; Maxwell, Jeffrey, and Lévesque 2011). More broadly, expertise is associated with faster information search, earlier termination of search, and reduced reliance on extensive information processing (Klein 2015). Taken together, this evidence suggests that experience may be associated not only with differences in the amount of information considered, but also with the speed, efficiency, and selectivity of information processing.

These insights provide a basis for examining how such experience-related differences may emerge in accelerator screening. Repeated exposure to comparable evaluation situations may enable evaluators to develop more differentiated knowledge about which signals merit attention and how those signals should be interpreted in context. Accordingly, evaluative experience may reshape the pace, focus, and cognitive character of screening by influencing how evaluators allocate attention, interpret application information, and deploy experience-informed judgment strategies. At the same time, these effects should not be understood as uniformly advantageous, because the value of learned cues depends on their fit with the decision environment. The present study therefore examines whether differences in evaluative experience are reflected in the pace, focus, and cognitive character of initial accelerator screening.

3. Data and methodology

3.1. Verbal protocol analysis

To investigate how evaluative experience is reflected in the cognitive processes of accelerator screening, this exploratory small-scale study employs Verbal Protocol Analysis (VPA), a method designed to capture information processing during task performance. VPA asks participants to "think aloud" while completing a task, producing a continuous verbalization of thoughts that reflect the contents of working memory at the moment of decision-making (Ericsson and Simon 1993). The method is based on the premise that information attended to during task performance is represented in working memory and can be verbalized when participants are instructed to report their thoughts concurrently with the task (Green 1998). Compared with retrospective methods such as surveys or interviews, which rely on participants' recollections after the fact, VPA captures decision processes as they unfold, reducing reliance on retrospective recall and post-hoc reconstruction (Ericsson and Simon 1993).

VPA has been extensively used to study investment decision-making among venture capitalists (Hall and Hofer 1993; Mason and Stark 2004) and business angels (T. Botelho and Mason 2024). These studies demonstrate the method's ability to reveal how investors attend to information, interpret signals, apply heuristics, and ultimately form evaluative judgments. By

capturing concurrent verbalizations during the evaluation task, VPA allows researchers to examine the sequence and content of cognitive operations underlying investment decisions, thereby overcoming many limitations associated with retrospective approaches (Harrison, Mason, and Smith 2015; Mason and Stark 2004). The method is particularly well suited for examining differences between novice and expert decision-makers, as it highlights variations in knowledge structures, attentional focus, and cognitive strategies (Harrison, Botelho, and Mason 2016). Moreover, the sequential nature of verbal protocols provides insight into how information is processed over time, making VPA especially valuable for analyzing complex, multi-stage tasks such as venture screening (Kuusela and Paul 2000).

Despite these advantages, VPA has several limitations. Participants may become self-conscious while verbalizing their thoughts, which could potentially influence their natural cognitive processes (Harrison, Mason, and Smith 2015). In addition, the controlled task environment may not fully replicate real-world evaluation conditions, and the coding and interpretation of verbal protocols can introduce subjectivity (Riquelme and Rickards 1992). Nevertheless, prior methodological research suggests that these limitations can be mitigated through careful study design. Ericsson and Simon (1993) argue that VPA is most appropriate when verbalizations reflect information currently in the participant's focus of attention, the task is sufficiently novel and not fully routinized, verbalization occurs concurrently with task performance rather than retrospectively, and participants receive clear instructions on how to report their thoughts. The present study followed these principles by asking evaluators to verbalize their thoughts while performing the screening task and by providing standardized instructions before each session.

3.2. Design of the startup screening task

The tasks were designed to closely replicate real-world startup screening situations faced by accelerator evaluators. This approach aligns with prior research emphasizing the importance of task realism for eliciting authentic cognitive processes during experimental decision-making tasks (Baldacchino et al. 2014). To enhance ecological validity, the researchers first observed startup screening sessions in several leading Iranian accelerators to understand how evaluators review and interpret applications in practice. Based on these observations, two representative startup applications were constructed: one in the health-tech domain and one in the ed-tech domain.

The applications were anonymized to protect confidentiality, while preserving the essential informational structure typically presented to accelerator evaluators. Each document was formatted using a standardized structure and ranged from approximately 750 to 900 words. The applications included consistent sections covering the problem statement, proposed solution, market opportunity, business model, competitive landscape, and founding team characteristics.

This structure mirrors the format commonly used in accelerator application forms and pitch summaries, allowing participants to engage with information similar to what they would encounter in practice. The selected length and level of detail were intended to create a task complex enough to generate sustained verbalization during the think-aloud protocol, while avoiding excessive cognitive load or participant fatigue. Together, the use of observed screening practices, standardized application structures, and realistic application content was intended to balance ecological validity with comparability across the screening tasks.

3.3. Sample selection

The study involved 10 accelerator evaluators, divided into two groups based on their level of screening experience: experienced and novice. Experienced evaluators had at least three years of experience working in startup accelerators and had screened a minimum of 300 startup applications, whereas novice evaluators had less than six months of screening experience and had reviewed fewer than 10 applications. These thresholds were intended to create a clear contrast between evaluators with substantial exposure to screening and those at the early stages of developing screening experience.

Given the exploratory and small-scale nature of the study, purposive sampling was used to identify participants with clearly differentiated levels of screening experience relevant to the research question. To reduce the possibility that differences in organizational maturity would confound individual screening experience, all participants were recruited from accelerators that had completed at least four acceleration cycles at the time of data collection. Consequently, variation in screening experience primarily reflected differences at the individual level, although organizational influences could not be completely ruled out. Two experienced–novice pairs were drawn from the same accelerator, while the remaining participants came from different accelerators. The intensive and detailed nature of VPA makes relatively small samples suitable for examining cognitive processes in depth rather than for statistical generalization (Trickett and Trafton 2009). Table 1 shows how the sample was selected to provide sufficiently differentiated experience levels for process-oriented comparison.

3.4. Data collection and analysis

The data collection involved face-to-face interviews with participants at the accelerators, followed by the "think-aloud" exercise. Evaluators were instructed to screen the two startup applications as they typically would, verbalizing their thoughts throughout the process. To minimize potential order or fatigue effects, the presentation order of the two applications was randomized across participants. Researchers intervened only when evaluators fell silent for extended periods, reminding them to continue verbalizing their thoughts. Each session was audio-recorded, yielding 20 verbal protocols, which were subsequently transcribed for analysis.

After transcription, the verbal protocols were segmented into discrete units, forming the fundamental units of analysis. The appropriate size, or "grain size", of a segment can

Table 1. Demographic information of evaluators.

Evaluator	Age	Gender	Education	Experience in accelerator	Screening experience
Experienced 1	40	M	M.A. in Entrepreneurship	3 Years	~300 applicants
Experienced 2	39	M	PhD. in Entrepreneurship	4 Years	~300 applicants
Experienced 3	43	M	MBA	5 Years	~500 applicants
Experienced 4	43	M	MBA	3 Years	~300 applicants
Experienced 5	35	M	MBA	3 Years	~400 applicants
Novice 1	26	M	M.A. in Entrepreneurship	3 Month	<10 applicants
Novice 2	31	M	M.A. in Management	2 Month	<10 applicants
Novice 3	29	M	MBA	3 Month	<10 applicants
Novice 4	28	M	MBA	6 Month	<10 applicants
Novice 5	35	M	M.A. in Management	4 Month	<10 applicants

vary according to the research question and may range from a phrase or sentence to a complete idea or chain of reasoning (Chi 1997). In this study, we followed the approach of Trickett and Trafton (2007) and defined a thought unit as a distinct, complete thought, typically expressed as a clause containing a subject and a verb. This level of segmentation allowed us to capture distinct evaluative thoughts while retaining sufficient contextual meaning. For example, "The revenue model needs further refinement" and "The technical team member has good work experience" were each treated as a separate thought unit.

Following segmentation, each thought unit was coded using an established framework for analyzing verbal protocols in investment decision-making (T. D. S. Botelho 2017; Harrison, Mason, and Smith 2015; Mason and Botelho 2016; Rogers and Mason 2010). Thought segments were classified into two dimensions: investment criteria (e.g., team, product, market, financials) and statement types (e.g., description, inference, recall). Tables 2 and 3 summarize these classifications.

To ensure coding reliability, two co-authors independently coded all verbal protocols using the coding manual, without discussing their coding until the independent coding process was complete. Krippendorff's alpha was 0.899 for screening criteria and 0.874 for statement type, indicating a high level of agreement between the coders. Discrepancies were subsequently reviewed and resolved through discussion and consensus.

In a separate analysis, the protocols were examined for evidence of heuristic reasoning, focusing on availability, representativeness, and anchoring and adjustment (Holcomb et al. 2009; Tversky and Kahneman 1974). Because heuristic reasoning could extend across

Table 2. Classification of thought segments in the protocols – screening criteria.

Screening criteria	Description
People	Issues regarding the entrepreneur, team, and inventor. Their background, experience, qualities, etc.
Product	The nature of the product: technical aspects, intellectual property protection, competitive advantages, design, etc.
Market	This includes points on the market: organization, growth, competition, geography, size, etc.
Pitch Deck	Specific comments on the pitch deck: length, presentation, content missing data, etc.
Exit	Who? When? How much? Type of exit. Existence of an exit plan
Financial Considerations	Amount of investment, amount raised, future funding needs, valuation, equity share, cash-flows, etc.
Accelerator Attributes	Issues regarding accelerator fit, including the accelerator experiences and interests.
Attributes of the Business	This includes a broad scope of issues: e.g., strategy, business model, risks, operations, time frame, etc.
Other	Comments on any aspects of the business which cannot be coded in any other category

Source: Adapted from Mason and Rogers (1997).

Table 3. Classification of thought segments in the protocols – statement type.

Statement type	Description
Description	Non-evaluative statement consisting of verbatim or paraphrased quotation of information presented in plan
Recall	Non-evaluative information obtained from past experience
Preconception	Judgmental statement based on previous experience/ background knowledge
Inference	Statement which involves a judgement on some part of the plan
Question	Statement which seeks further information
Action	Statement of intention or action to be performed (for example, to search for a source of information)
Comment	Uncodable or irrelevant statement

Source: Mason and Rogers (1997).

multiple consecutive thought units, it was identified at the level of reasoning episodes. Representativeness was identified when features of the founding team, market, product, or business model were interpreted in relation to a broader pattern or prototype developed from prior knowledge or experience (Holcomb et al. 2009). Availability was identified when a remembered startup, investment experience, market event, or other personally encountered case was used as a basis for evaluating the venture under review. Anchoring and adjustment was identified when an initial impression, salient application feature, preliminary assessment of attractiveness or risk, or previously established expectation served as a reference point that was subsequently adjusted in light of additional information (Harrison, Mason, and Smith 2015; Tversky and Kahneman 1974). These heuristic episodes were examined separately from the screening-criterion and statement-type coding and subsequently compared across novice and experienced evaluators.

4. Results

Before examining how novice and experienced evaluators processed information, we first establish that both groups engaged meaningfully with the screening task and reached final decisions. Table 4 summarizes the screening decisions for the two applications by evaluators. As shown in the table, the final acceptance decisions were broadly similar across the two groups, with only three divergent decisions across the ten evaluations. Despite this convergence in screening outcomes, the verbal protocols revealed differences in the processes through which these decisions were reached. The following sections examine these differences across four dimensions of the screening process: decision time, evaluation criteria, statement types, and heuristic use.

Notable differences emerged in decision time. To assess whether these patterns were specific to one application, we compared novice and experienced evaluators separately for Applications A and B and then examined the aggregate patterns. The difference was evident across both applications and in the combined results. Experienced evaluators required an average of 306 seconds compared with 458 seconds for novices when evaluating Application A, and 315 seconds compared with 511 seconds for Application B. Across both applications, average screening times were 310.5 seconds for experienced evaluators and 484.5 seconds for novices. Notably, the longest screening time recorded among experienced evaluators (360 seconds) remained shorter than the shortest screening time among novices (385 seconds), indicating a clear separation between the two groups. Overall, experienced evaluators completed the screening task substantially faster than novice evaluators, suggesting greater confidence in their decision-making process.

Table 4. Decision and length of time to make screening decision.

	Application A		Application B			Application A		Application B	
Evaluator	Decision	Time (s)	Decision	Time (s)	Evaluator	Decision	Time (s)	Decision	Time (s)
Exp-1	✓	310	✓	315	Nov-1	✓	385	×	505
Exp-2	×	215	✓	250	Nov-2	✓	410	✓	460
Exp-3	✓	340	✓	340	Nov-3	×	610	✓	560
Exp-4	✓	360	✓	350	Nov-4	✓	485	✓	440
Exp-5	✓	305	✓	320	Nov-5	✓	400	✓	590
Average	–	**306**	–	**315**	**Average**	–	**458**	–	**511**

Key: × = reject/✓ = proceed to interview; Note: All time measures are reported in seconds.

Analysis of evaluation criteria revealed systematic differences in emphasis between novice and experienced evaluators (Table 5). Novices primarily focused primarily on "people", "product", and "financial considerations", which together accounted for approximately 73% of their thought units. In contrast, experienced evaluators emphasized "people", "market", and "accelerator attributes" -such as strategic fit and alignment- which comprised approximately 80% of their statements.

Both groups assigned high importance to team-related factors, indicating a shared recognition of the central role of founding teams in early-stage ventures. Differences emerged, however, in their secondary priorities. Novices frequently referred to information presented in the application materials, including product features, development plans, intellectual property considerations, and financial indicators such as projected revenues and commission structures. For example, one novice evaluator noted, "10% of each visit is very low", reflecting attention to a financial parameter provided in the proposal. Similarly, novices often discussed stated market size estimates and timelines for product delivery.

In contrast, experienced evaluators more frequently referred to market conditions and accelerator-related considerations that were not explicitly detailed in the applications. When evaluating application A, an experienced evaluator stated, "this team does not align with our accelerator's policies". When assessing application B, another commented, "despite nearly four years of startup activity in this market, no venture has yet achieved a strong market position". These statements show that experienced evaluators drew on information extending beyond the application materials, including prior market knowledge and familiarity with accelerator operations.

Analysis of statement types (Table 6) further illustrates differences in how novice and experienced evaluators engaged with application materials. Both groups made extensive use of inference statements, reflecting interpretive and judgment-oriented reasoning during the screening process. However, the balance of other statement types diverged systematically.

Among novices, description and question statements were the next most frequent categories. These included direct references to application content and requests for additional information. For example, one novice asked, "has the product received the necessary licenses?" while another noted descriptively, "the market size is estimated at

Table 5. Verbal protocol frequency counts by screening criterion.

	% of thought units by type (figure in brackets is ranking)					
	Application A		Application B		Overall	
Screening criterion	Experienced	Novice	Experienced	Novice	Experienced	Novice
People	29.4 (1)	30.2 (1)	29.0 (1)	35.0 (1)	29.2 (1)	32.5 (1)
Market	26.5 (2)	11.1 (4)	29.0 (1)	10.0 (4)	27.7 (2)	10.6 (4)
Accelerator Attributes	23.5 (3)	3.2 (6)	22.6 (2)	0.0 (7)	23.1 (3)	1.6 (7)
Product	11.8 (4)	22.2 (2)	9.7 (3)	25.0 (2)	10.8 (4)	23.6 (2)
Financial Considerations	5.9 (5)	20.6 (3)	3.2 (5)	13.3 (3)	4.6 (5)	17.1 (3)
Attributes of the Business	2.9 (6)	3.2 (6)	6.5 (4)	10.0 (4)	4.6 (5)	6.5 (5)
Pitch Deck	0.0 (7)	3.2 (6)	0.0 (6)	3.3 (5)	0.0 (6)	3.25 (6)
Exit	0.0 (7)	4.8 (5)	0.0 (6)	1.7 (6)	0.0 (6)	3.25 (6)
Other	0.0 (7)	1.6 (7)	0.0 (6)	1.7 (6)	0.0 (6)	1.6 (7)

Note: Spearman's Rho rank order correlation: Novice A-Novice B ($\rho = 0.87$); Experienced A-Experienced B ($\rho = 0.97$); Novice-Experienced A ($\rho = 0.66$); Novice-Experienced B ($\rho = 0.46$).

Table 6. Verbal protocol frequency counts by statement type.

	% of thought units by type (figure in brackets is ranking)					
	Application A		Application B		Overall	
Statement type	Experienced	Novice	Experienced	Novice	Experienced	Novice
Inference	44.1% (1)	50.7% (1)	48.4% (1)	42.6% (1)	46.2% (1)	47.5% (1)
Preconception	14.7% (2)	2.8% (5)	25.8% (2)	2.1% (5)	20.0% (2)	2.5% (5)
Description	14.7% (2)	22.5% (2)	16.2% (3)	25.5% (2)	15.4% (3)	23.7% (2)
Action	5.9% (4)	4.2% (4)	3.2% (4)	2.1% (5)	4.6% (5)	3.4% (4)
Recall	11.8% (3)	4.2% (4)	3.2% (4)	12.8% (4)	7.7% (4)	7.6% (4)
Question	5.9% (4)	12.7% (3)	3.2% (4)	14.9% (3)	4.6% (5)	13.6% (3)
Comment	2.9% (5)	2.8% (5)	0.0% (5)	0.0% (6)	1.5% (6)	1.7% (6)

Note: Spearman's Rho rank order correlation: Novice A-Novice B ($\rho = 0.95$); Experienced A-Experienced B ($\rho = 0.95$); Novice-Experienced A ($\rho = 0.57$); Novice-Experienced B ($\rho = 0.71$).

5 million potential users". Similarly, in evaluating the other application, a novice remarked, "they did not specify how the product differs from competitors". These statements show that novice evaluators frequently referred directly to information contained in the applications and identified information gaps during screening.

Experienced evaluators, by contrast, more frequently used what we term preconception statements- references to prior knowledge, sector experience, or familiar patterns that extended beyond the application text. These appeared alongside descriptive statements but drew on domain-specific knowledge rather than application content alone. For instance, regarding application A, one evaluator stated, "the online language-learning market in Iran is challenging due to intense international competition". Assessing application B, another noted, "the healthcare sector is restrictive due to licensing issues, and regulatory bodies often complicate implementation". These statements show that experienced evaluators incorporated prior sector knowledge into their screening judgments alongside information contained in the applications.

Our analysis also revealed indicative differences in how novice and experienced evaluators appeared to use heuristics. Following Holcomb et al. (2009), the analysis focused on three heuristics originally proposed by Tversky and Kahneman (1974): availability, representativeness, and anchoring and adjustment. While both groups showed evidence consistent with heuristic reasoning, the sources of these judgments appeared to differ across experience levels. Clearer evidence was observed for availability and representativeness, whereas convincing evidence of anchoring and adjustment was limited and not consistently observed across the protocols.

Novice evaluators more often appeared to draw heuristic judgments from information contained in the application materials and personally salient experiences. Availability-based reasoning was evident when personally accessible experiences were used to interpret the focal venture. For example, one novice evaluator stated, "I actually had a similar problem just a few days ago, and a product like this could be really helpful . . . ". Representativeness-based reasoning was also observed when evaluators compared a venture with broad or generalized prototypes. For example, one novice stated, "successful businesses usually have a large market, and this business has exactly that characteristic". Such reasoning linked visible characteristics of the focal venture to readily accessible experiences or generalized notions of what a successful venture should look like.

Experienced evaluators, in contrast, more often appeared to draw on prior screening cases and accumulated domain knowledge when forming heuristic judgments. Availability-based reasoning involved the retrieval of prior ventures or market experiences. For example, one experienced evaluator stated, "recently, I have seen two or three teams getting into this market . . . this market is going to grow soon". Representativeness-based reasoning involved comparisons with more context-specific or industry-related prototypes developed through prior experience. For example, an experienced evaluator stated, "this team is a bit too technical . . . they don't have anyone for marketing . . . teams like this usually have a kind of product-focused mindset rather than a business-oriented one". This episode reflects a generalized prototype of technically oriented teams derived from prior experience and applied to the focal venture.

Overall, the protocols suggest an experience-related difference in the informational basis of heuristic reasoning: novices more often drew on information contained in the applications and personally salient experiences, whereas experienced evaluators more often drew on prior screening cases and accumulated domain knowledge. Given the exploratory nature of the analysis, these patterns are interpreted as indicative differences in heuristic reasoning rather than definitive evidence that evaluators consistently employed specific heuristics.

Taken together, these results reveal a consistent pattern of differences between novice and experienced evaluators. Although both groups reached similar screening outcomes, they differed consistently across multiple dimensions of cognitive processing: decision time, attentional focus, statement types, and heuristic reasoning. Experienced evaluators completed the screening task more quickly and appeared to attend to a more selective set of evaluation cues than novice evaluators. However, the broadly similar screening decisions do not indicate that these differences in cognitive processing translated into superior screening judgments. The following section discusses the theoretical and practical implications of these findings.

5. Discussion

Our research set out to examine how novice and experienced evaluators differ in the cognitive processes underlying accelerator screening. Using verbal protocol analysis, we captured the real-time thinking of both groups as they assessed startup applications. The results reveal that while novice and experienced evaluators reached broadly similar decision outcomes, they diverged in decision time, evaluative priorities, and interpretive strategies.

The most immediately visible difference between groups was decision time. Experienced evaluators consistently reached conclusions more quickly than novices, indicating a substantial difference in the pace of screening. When considered alongside the other dimensions of decision process, this difference is consistent with the development of more selective and structured information processing. Through repeated exposure to venture evaluations, experienced evaluators appear to have developed pattern recognition capabilities that allow them to interpret complex and ambiguous information more selectively. Similar dynamics have been observed in studies of angel investors and venture capitalists, where experience is associated with faster judgments under uncertainty (Hall and Hofer 1993; Harrison, Mason, and Smith 2015). From an experiential

learning perspective, this pattern is consistent with the accumulation and internalization of evaluative knowledge through repeated experience and reflection. The observed time differences suggest that experience may enable evaluators to reach screening decisions with less extensive information processing.

Analysis of evaluation criteria revealed systematic differences in what novice and experienced evaluators attended to. While both groups emphasized the importance of the founding team, novice evaluators devoted substantial attention to product features and financial indicators. These criteria are explicitly presented in the application materials and provide novices with a sense of analytical rigor and objectivity. However, such indicators are often difficult to assess reliably at early stages and may require substantial verification. Experienced evaluators, by contrast, focused more strongly on market plausibility and accelerator fit. The difference was particularly pronounced for these criteria: market plausibility ranked second among experienced evaluators but only fourth among novices, while accelerator-related considerations ranked third among experienced evaluators but seventh among novices. These dimensions are closely anchored in external context and organizational experience and can be interpreted using accumulated knowledge. At the screening stage, evaluators appear primarily concerned with whether a venture warrants further attention and deeper evaluation. Rather than attempting to fully assess product feasibility or financial sustainability, which often requires additional data and due diligence, experienced evaluators concentrate on signals that can be interpreted reliably using external knowledge and prior experience. Team competence, market plausibility, and accelerator fit may therefore serve as informative screening cues in the context examined here. Consistent with prior research, early-stage screening is often oriented toward eliminating clearly unpromising ventures in order to allocate limited resources more effectively (Harrison, Mason, and Smith 2015; Maxwell, Jeffrey, and Lévesque 2011; Yin and Luo 2018).

This pattern is consistent with a selective narrowing of attention rather than a comprehensive assessment of every feature. By concentrating cognitive effort on a limited set of cues, experienced evaluators may reduce the amount of information that needs to be processed during the initial screening stage.

Novice evaluators more often grounded their judgments in information explicitly provided in the applications, including product descriptions, financial projections, and development plans. Their frequent use of descriptive and questioning statements reflects an effort to manage uncertainty through detailed examination of available information. Experienced evaluators, by contrast, more frequently drew on preconceptions, prior knowledge, and contextual interpretations involving market dynamics, regulatory constraints, and organizational capabilities. Rather than treating application materials as sufficient for forming a judgment, experienced evaluators relied more strongly on accumulated industry knowledge, contextual understanding, and prior screening experience. This difference suggests that experience may shift uncertainty management from extensive examination of application content toward greater reliance on contextual interpretation.

This shift in information reliance is consistent with the development of internalized evaluative knowledge structures. Through repeated exposure to venture cases and outcomes, experienced evaluators may develop cognitive representations of recurring success and failure patterns that guide the interpretation of new cases. Applications are therefore evaluated not only in relation to the information they contain, but also in relation to prior

cases and accumulated knowledge. This finding aligns with prior research showing that expertise can increase reliance on situational knowledge and pattern recognition in complex decision environments (Baron and Ensley 2006; Bernstein, Korteweg, and Laws 2017; Huang and Pearce 2015).

Differences in heuristic use further support this interpretation. Although both novice and experienced evaluators relied on availability and representativeness heuristics, the content of these heuristics differed. Novices tended to draw on personal or socially shared experiences, resulting in broader and more generalized prototypes of promising ventures. Experienced evaluators, in contrast, relied more on heuristics grounded in accumulated screening histories and domain-specific cases. Their judgments drew on remembered examples of regulatory barriers, commercialization failures, and team-related challenges, as well as on experience-shaped cognitive schemas. This pattern is consistent with prior work suggesting that heuristics can function as adaptive strategies for simplifying judgment under uncertainty, while their content and usefulness depend on the knowledge available to the decision-maker and the context in which the heuristic is applied (Bingham and Eisenhardt 2011; Gigerenzer, Reb, and Luan 2022; Harrison, Mason, and Smith 2015; Maxwell, Jeffrey, and Lévesque 2011).

Taken together, the findings suggest that experience is associated with meaningful changes in both the efficiency and structure of screening cognition. Experienced evaluators reached decisions more quickly and relied on a narrower and more selective set of cues, drawing more heavily on contextual knowledge and prior experience. However, these differences in cognitive processing were not accompanied by substantial differences in screening outcomes. The convergence of outcomes alongside divergent cognitive processes suggests that experience may change the route through which evaluators arrive at screening judgments without necessarily changing the immediate decision to advance or reject an application.

One possible interpretation is that different cognitive paths can converge on similar screening decisions: novices may arrive at a decision through more extensive engagement with application-based information, whereas experienced evaluators may reach a similar conclusion through selective attention to contextual cues and experience-based heuristics. A second possibility is that the screening task itself constrained outcome variation, such that the two applications did not provide sufficient differentiation to produce stronger divergence between groups. The present study cannot distinguish conclusively between these explanations. The similarity in outcomes should therefore not be interpreted as evidence that experience has no effect on evaluative quality, nor as evidence that experienced evaluators make better decisions. Instead, it suggests that experience may primarily reconfigure how screening judgments are reached. In this sense, experience appears to function as an adaptive response to the uncertainty and information constraints of initial accelerator screening, while the usefulness of particular cues remains dependent on the decision context.

5.1. Implications for theory

This study contributes to the literature on accelerator selection by shifting attention from evaluation criteria to the cognitive processes underlying early-stage screening decisions. While prior research has primarily examined what evaluators look for, our findings suggest

that how evaluators process information varies systematically with experience, particularly under conditions of high uncertainty and time pressure. In doing so, the study reinforces the view that screening is not merely a simplified version of later-stage investment evaluation, but a distinct decision context with unique cognitive demands, where evaluators must make consequential judgments under substantial uncertainty, limited information, and time constraints. The findings therefore position accelerator screening as a setting in which evaluative expertise is expressed through selective information processing rather than simply through the accumulation of additional knowledge.

Second, the study advances understanding of learning from experience in entrepreneurial finance by showing how repeated exposure to screening tasks may reshape evaluative cognition. Consistent with experiential learning theory, experience does not simply accumulate knowledge but may transform the way evaluators attend to information, interpret signals, and make judgments in real time. The findings suggest that experienced evaluators rely less on exhaustive analysis of application content and more on internally developed cognitive schemas and contextual interpretation, indicating that learning may manifest in changes in processing efficiency and selectivity rather than necessarily in greater judgment accuracy. This extends prior research on venture capitalists and angel investors to the accelerator context, where such dynamics have remained underexplored.

The findings also point to a situated dimension of evaluative expertise. Experienced evaluators placed greater emphasis on market plausibility and accelerator fit, suggesting that the cues regarded as informative are not necessarily universal but may depend on the venture, industry, accelerator mission, and broader ecosystem context. In particular, accelerator fit reflects knowledge embedded in the organizational environment, including its objectives, resources, and accumulated experience with ventures. Expertise in accelerator screening may therefore involve not only individual knowledge structures but also the ability to interpret venture signals in relation to the specific organizational and institutional context in which evaluation occurs. This extends the understanding of expertise beyond an individual-level stock of experience toward a more context-sensitive conception of evaluative capability.

Finally, the study contributes to the heuristics literature by showing that heuristic use may differ in its informational basis with experience in early-stage screening contexts. Both novice and experienced evaluators rely on heuristics; however, their nature and sources differ. Novices apply generalized, application-driven heuristics, whereas experienced evaluators draw on case-based, domain-specific heuristics grounded in prior screening encounters. This pattern is consistent with the view that heuristics can function as adaptive strategies for simplifying judgment under uncertainty, while their diagnostic value depends on the knowledge and context from which they are constructed (Bingham and Eisenhardt 2011; Gigerenzer, Reb, and Luan 2022; Harrison, Mason, and Smith 2015; Maxwell, Jeffrey, and Lévesque 2011). Thus, rather than suggesting that experience simply increases heuristic reliance, the findings indicate that experience may alter which information is incorporated into heuristic judgments and how those heuristics are applied. In this way, the study offers a more nuanced understanding of how evaluative expertise may develop in accelerator settings and, more broadly, how decision-making adapts to conditions of uncertainty in early-stage entrepreneurship.

5.2. Implications for practice

Beyond its theoretical contributions, this study offers practical implications for accelerator managers, evaluators, and program designers seeking to improve the efficiency and consistency of selection processes.

First, the findings suggest that evaluator experience can be a valuable organizational resource in high-volume screening contexts. Experienced evaluators reached screening decisions more quickly and relied on a more selective set of cues, including team characteristics, market plausibility, and accelerator fit. Accelerators may therefore benefit from involving experienced evaluators in initial screening and using their expertise to support rapid prioritization of applications.

Second, the results highlight the importance of developing evaluative capabilities among novice evaluators. Accelerators can support the development of evaluative expertise through structured training, mentoring, joint evaluation sessions and post-decision debriefings. Making experienced evaluators' reasoning more explicit through case-based discussions or think-aloud exercises may help novices develop greater sensitivity to contextual cues without simply reproducing expert judgments.

Finally, the findings suggest that screening frameworks should reflect the informational constraints of the initial stage. Rather than attempting comprehensive evaluation, accelerators may benefit from prioritizing a limited set of contextually relevant cues and reserving more detailed assessment for later stages. This approach may help allocate evaluative resources more efficiently while preserving the developmental purpose of accelerator selection.

5.3. Limitations and future research directions

This study has several limitations that also point to directions for future research. First, the exploratory design and small sample limit the extent to which the observed patterns can be generalized across evaluator populations. Although VPA provides rich process-level data, the study was designed to examine cognitive differences in depth rather than to establish statistically generalizable effects. The all-male composition of the sample further limits the diversity of evaluator perspectives represented in the study. Future research could therefore examine larger and more diverse samples, including greater gender diversity, to assess the robustness and boundary conditions of the observed patterns.

Second, the study is situated within the Iranian accelerator ecosystem, and the findings may partly reflect its institutional, regulatory, and market conditions. Such conditions can shape which cues evaluators regard as informative and how they interpret market plausibility, regulatory constraints, and venture potential. More broadly, the diagnostic value of particular cues may vary across industries and institutional environments. Future research could compare evaluators across different institutional, industry, and ecosystem contexts to examine how experience interacts with different decision environments.

Organizational context represents a related boundary condition. Although participants were recruited from established accelerators and both experience groups were represented across organizations, the study did not systematically control for accelerator-specific routines, missions, or socialization processes. Some aspects of screening may therefore reflect not only individual evaluative experience but also the organizational context in which that experience is developed, particularly when evaluators assess accelerator fit. Future research could examine

how accelerator mission, program design, and accumulated organizational knowledge shape screening cognition and the development of evaluative expertise. This may be particularly relevant for accelerators supporting impact-oriented ventures, where evaluation involves balancing commercial potential with social or environmental objectives (Laspia et al. 2025; Viglialoro et al. 2025; Viglialoro, Seguel, and Landoni 2024). Comparing novice and experienced evaluators in such settings could reveal whether experience shapes the cognitive strategies used to assess these multiple objectives and further clarify how evaluative expertise develops across different accelerator contexts.

Third, the analysis was limited to two startup applications and the initial screening stage. Although the applications represented different venture domains, the small number of cases means that some observed differences may reflect case-specific characteristics. Moreover, cognitive strategies may change as evaluators move from initial screening to later-stage due diligence and investment decisions. Future research could therefore use larger sets of applications across industries and longitudinal designs spanning multiple selection stages to examine how evaluative strategies develop and change over time.

Finally, VPA captures information that evaluators consciously verbalize during task performance and therefore cannot fully reveal implicit or automatic cognitive processes. Complementary methods, such as eye-tracking, behavioral experiments, or retrospective interviews, could be combined with VPA to examine cognitive processes that may not be verbalized. Such multimethod approaches could further clarify the mechanisms underlying the faster and more selective processing observed among experienced evaluators.

6. Conclusion

This study examined how evaluative experience shapes the cognitive processes involved in the initial screening of startup accelerator applications. The findings indicate that expertise is reflected not simply in accumulated knowledge or different screening outcomes, but in a different way of navigating the uncertainty and information constraints of early-stage evaluation. Experienced evaluators appeared to have developed more efficient and selective screening routines, drawing on contextual knowledge and prior cases to interpret a narrower set of cues, whereas novices engaged more extensively with the information contained in applications. At the same time, the broadly similar screening outcomes caution against equating faster or more selective processing with superior judgment. Instead, the findings suggest that evaluative expertise may be understood as a situated capability for filtering and interpreting information, shaped by repeated experience and, potentially, by the organizational context in which that experience is developed.

By opening the cognitive processes underlying accelerator screening to direct observation through VPA, this study shifts attention from the criteria evaluators use to the ways those criteria and other signals are processed in real time. This perspective highlights experiential learning as a process through which evaluators develop more context-sensitive approaches to navigating ambiguous information. For accelerator practice, the implication is not simply to rely more heavily on experienced evaluators, but to create opportunities through which novice evaluators can develop the ability to recognize and interpret relevant cues in context. More broadly, the study provides an exploratory foundation for future research examining how evaluative expertise develops across different accelerator missions, industries, and institutional environments.

Author contributions

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